UNIVERSITY OF WATERLOO

SYDE 556/750 – Simulating Neurobiological Systems

# Practice Test 3B: Nengo & Simple Dynamics

Student ID: _____ Time: 30 minutes Total: 20 marks (plus 2 bonus)

- **Closed book.** No notes, textbook, calculator, phone, or other devices.
- The marks for each question are shown in brackets on the right.
- Write your answers in the space provided. Show your work — partial credit is given for correct reasoning.
- Unless stated otherwise, every synapse is the first-order low-pass $h(t) = \frac{1}{\tau} e^{-t/\tau}$.

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**Question 1.** **[4 marks]**

(a) A Nengo ensemble represents $\mathbf{x} \in \mathbb{R}^2$ with `radius=2`. One of its LIF neurons has encoder $e = (0, 1)$, a (normalized) intercept of $-0.5$, and a maximum rate of 150 Hz. In the left box, sketch this neuron's firing rate along the line $\mathbf{x} = (0, s)$, and along the line $\mathbf{x} = (s, 0)$, for $s \in [-2, 2]$. Label where it starts firing.

(b) The same kind of ensemble (now 1-D, radius 1) is driven by $x(t) = \sin(2\pi \cdot 2t)$. Its decoded output is recorded with three probes: `synapse=0.001`, `synapse=0.01` and `synapse=0.1`. In the right box, sketch what each probe records over one second. Does changing the probe synapse change what the ensemble represents?

| (a) | (b) |
|---|---|

**Question 2.** **[4 marks]**

A population representing $x$ receives an input $u$ through a connection computing $g(u)$, and has a recurrent connection computing $f(x)$. Both connections use a synapse with time constant $\tau$. The represented value obeys

$$\tau \frac{dx}{dt} = g(u) + f(x) - x.$$

(a) Explain what each term represents, and where the $-x$ comes from. What would $x$ do if the recurrent connection were removed?

(b) Using $\tau = 50$ ms, find the recurrent and input transforms so that the network implements the low-pass filter

$$\frac{dx}{dt} = -\frac{1}{0.5}\, x + \frac{1}{0.5}\, u.$$

Draw the network (populations, connections, transforms). What time constant would you see in the response to a step in $u$?

**Question 3.** **[4 marks]**

A population with radius 1 and synaptic time constant $\tau$ has input connection $g(u) = u$ and recurrent connection $f(x) = x - x^2$. The input is held constant.

(a) Write the resulting dynamics $\tau\dot{x} = \varphi(x)$. For $u = 0.25$, find all equilibria and state whether each is stable or unstable.

(b) Sketch $\varphi(x)$ for $u = 0.25$ with arrows on the $x$-axis. Describe what $x$ does starting from $x(0) = 0$ and from $x(0) = -0.8$. What changes if the input is lowered to $u = -0.1$?

**Question 4.** **[4 marks]**

You want a 2-D population to oscillate at 1 Hz:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} \mathbf{x}, \qquad \omega = 2\pi \text{ rad/s}.$$

(a) The recurrent synapse is $\tau = 0.1$ s. What recurrent transform $A'$ do you use? (Leave $\pi$ in your answer.)

(b) A brief input kick leaves the state at $\mathbf{x}(0) = (0, 0.8)$. In the box, sketch *all of the following*:

- $x_1(t)$ and $x_2(t)$ on the same time axis for $0 \le t \le 2$ s;
- the trajectory in the $(x_1, x_2)$ plane, with an arrow showing its direction.

(c) Why does the network need the kick? In a real Nengo simulation, what sets the amplitude it ends up oscillating at?

**Question 5.** **[4 marks]**

An integrator $\dot{x} = u$ is built with recurrent transform 1 and recurrent synapse $\tau = 0.1$ s. It starts at $x(0) = 0$ and receives a ramp input:

$$u(t) = \begin{cases} 4t, & 0 \le t < 0.5 \text{ s}, \\ 0, & t \ge 0.5 \text{ s}. \end{cases}$$

(a) Find the ideal $x(t)$ for all $t \ge 0$ and sketch it.

(b) The input connection uses transform $\tau$ but a *faster* synapse, $\tau_{\text{in}} = 10$ ms. Sketch how the decoded $x(t)$ differs from your answer to (a). Roughly how large is the difference just before $t = 0.5$ s, and what happens after the ramp stops?

**Bonus Question 1.** **[2 bonus marks]**

You build the integrator of Question 5 (with matched synapses) and stop the input at $t = 0.5$ s. Over the next few seconds, the decoded value creeps away from 0.5. In plain language, why does a neural integrator drift? With spiking neurons the drift looks jittery and partly random, but with the same neurons in rate mode (`LIFRate`) it is smooth and predictable. Why?